

```
import streamlit as st

from googletrans import Translator

import speech_recognition as sr

from gtts import gTTS

import os

import PyPDF2 # For PDF file handling

from io import StringIO # For text file handling


# Initialize translator

translator = Translator()


# Language mapping (full names to language codes)

LANGUAGES = {

    "English": "en",

    "Spanish": "es",

    "French": "fr",

    "German": "de",

    "Chinese (Simplified)": "zh-cn",

    "Japanese": "ja",

    "Hindi": "hi",

    "Arabic": "ar",

    # Indian Languages

    "Bengali": "bn",

    "Tamil": "ta",

    "Telugu": "te",

    "Marathi": "mr",

    "Gujarati": "gu",

    "Kannada": "kn",

    "Malayalam": "ml",

    "Punjabi": "pa",

    "Odia": "or",
```

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"Urdu": "ur",  
"Sanskrit": "sa",  
}
```

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# Function to convert text to speech
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def text_to_speech(text, lang_code):  
    tts = gTTS(text=text, lang=lang_code, slow=False)  
    tts.save("translated_audio.mp3")  
    audio_file = open("translated_audio.mp3", "rb")  
    audio_bytes = audio_file.read()  
    st.audio(audio_bytes, format="audio/mp3")  
    os.remove("translated_audio.mp3") # Clean up the file
```

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# Function to capture voice input
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def capture_voice_input():  
    recognizer = sr.Recognizer()  
    with sr.Microphone() as source:  
        st.write("Speak now...")  
        audio = recognizer.listen(source)  
        try:  
            text = recognizer.recognize_google(audio)  
            return text  
        except sr.UnknownValueError:  
            st.error("Sorry, I could not understand the audio.")  
        except sr.RequestError:  
            st.error("Sorry, there was an issue with the speech recognition service.")  
    return None
```

```
# Function to extract text from a PDF file
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def extract_text_from_pdf(file):  
    pdf_reader = PyPDF2.PdfReader(file)
```

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text = ""

for page in pdf_reader.pages:
    text += page.extract_text()

return text


# Function to extract text from a text file
def extract_text_from_txt(file):
    return file.read().decode("utf-8")


# Streamlit app
st.title("TransLingua - Text, Voice, and Document Translation Tool")

st.write("""
Translate text, voice input, or entire documents into multiple languages seamlessly.

Upload a document, input text, or speak, select the target language, and get the translation with
audio output.
""")


# Input method selection
input_option = st.radio("Choose input method:", ("Text", "Voice", "Document"))


input_text = ""

if input_option == "Text":
    input_text = st.text_area("Enter text to translate:", height=200)
elif input_option == "Voice":
    if st.button("Start Voice Input"):
        input_text = capture_voice_input()
    if input_text:
        st.write(f"*You said:* {input_text}")
else: # Document input
    uploaded_file = st.file_uploader("Upload a document (PDF or TXT):", type=["pdf", "txt"])
    if uploaded_file:

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if uploaded_file.type == "application/pdf":
    input_text = extract_text_from_pdf(uploaded_file)
elif uploaded_file.type == "text/plain":
    input_text = extract_text_from_txt(uploaded_file)
st.write("*Extracted Text:*")
st.info(input_text)

# Target language selection
target_lang = st.selectbox("Select target language:", list(LANGUAGES.keys()))

# Translate button
if st.button("Translate"):
    if input_text:
        try:
            # Detect source language
            detected_lang = translator.detect(input_text).lang
            source_lang_name = [k for k, v in LANGUAGES.items() if v == detected_lang][0]

            # Perform translation
            translation = translator.translate(
                input_text,
                src=detected_lang, # Automatically detect source language
                dest=LANGUAGES[target_lang] # Use language code for target
            )

            # Display results
            st.subheader("Translation Results:")
            col1, col2 = st.columns(2)
            with col1:
                st.write(f"*Detected Source Language:* {source_lang_name}")
                st.info(input_text)

```

```
with col2:

    st.write(f"*Translated Text ({target_lang}):*")

    st.success(translation.text)


# Convert translated text to speech
st.write("---")

st.write("*Listen to the translation:*")

text_to_speech(translation.text, LANGUAGES[target_lang])

except Exception as e:

    st.error(f"An error occurred during translation: {e}")

else:

    st.warning("Please provide input text, use voice input, or upload a document.")


# Footer
st.write("---")

st.write("© 2023 TransLingua. Powered by Streamlit, Google Translate API, and gTTS.")
```